Chat Link: https://parley.mit.edu/share/feOxZ3vhcbSY9p2Fbiiik
https://parley.mit.edu/share/feOxZ3vhcbSY9p2Fbiiik

```
global_data = {
  "attribute_1":        "total_cost",
  "attribute_2":        "material_continuity",
  "attribute_3":        "power_density",
  "attribute_4":        "thermal_performance",
  "attribute_5":        "manufacturability",

  "direction_attribute_1":  "negative",   # lower cost  → higher utility
  "direction_attribute_2":  "positive",   # higher score → higher utility
  "direction_attribute_3":  "positive",
  "direction_attribute_4":  "positive",
  "direction_attribute_5":  "positive",

  # Cost bounds: floor = tiny Al scooter winding (India); ceiling = heavy Cu truck motor (EU/US)
  "attribute_1_min":      5.00,
  "attribute_1_max":     500.00,
  # Score bounds: absolute extremes of 1–10 scoring rubric
  "attribute_2_min":      1.0,
  "attribute_2_max":      10.0,
  "attribute_3_min":      1.0,
  "attribute_3_max":      10.0,
  "attribute_4_min":      1.0,
  "attribute_4_max":      10.0,
  "attribute_5_min":      1.0,
  "attribute_5_max":      10.0,

  "copper_price_per_ton":   9500,       # USD/ton → $9.50/kg
  "aluminum_price_per_ton":  2500,       # USD/ton → $2.50/kg
  "copper_price_per_kg":   9.5,       # USD/ton → $9.50/kg
  "aluminum_price_per_kg":  2.5,       # USD/ton → $2.50/kg
}

regional_data = [
  {
    "region":                 "United States",
    "copper_product_market_share":    0.97,  # near-universal Cu incumbent
    "aluminum_product_market_share":   0.03,
    "weight_attribute_1": 0.20,  # total_cost       — moderate; performance-led market
    "weight_attribute_2": 0.20,  # material_continuity — supply chain discipline
```

```
        "weight_attribute_3": 0.25,   # power_density      — Tesla/GM/Rivian compete on kW/kg
        "weight_attribute_4": 0.25,   # thermal_performance — sustained power rating critical
        "weight_attribute_5": 0.10,   # manufacturability  — Cu hairpin mature; less decisive
    },
    {
        "region":                       "China",
        "copper_product_market_share":      0.78,   # largest Al adoption globally; still Cu-majority
        "aluminum_product_market_share":    0.22,
        "weight_attribute_1": 0.35,   # total_cost         — dominant; hyper-competitive EV pricing
        "weight_attribute_2": 0.20,   # material_continuity
        "weight_attribute_3": 0.20,   # power_density
        "weight_attribute_4": 0.15,   # thermal_performance — lower-tier EVs less demanding
        "weight_attribute_5": 0.10,   # manufacturability
    },
    {
        "region":                       "India",
        "copper_product_market_share":      0.92,
        "aluminum_product_market_share":    0.08,
        "weight_attribute_1": 0.40,   # total_cost         — affordability-first; highest single weight
        "weight_attribute_2": 0.15,   # material_continuity
        "weight_attribute_3": 0.15,   # power_density      — 2W/
        "weight_attribute_4": 0.15,   # thermal_performance
        "weight_attribute_5": 0.15,   # manufacturability  — Al welding infrastructure limited
    },
    {
        "region":                       "Europe",
        "copper_product_market_share":      0.96,
        "aluminum_product_market_share":    0.04,
        "weight_attribute_1": 0.15,   # total_cost         — lowest priority; premium OEM market
        "weight_attribute_2": 0.20,   # material_continuity — CBAM + critical minerals policy
        "weight_attribute_3": 0.30,   # power_density      — kW/kg is brand differentiator
        "weight_attribute_4": 0.25,   # thermal_performance — EU homologation stringency
        "weight_attribute_5": 0.10,   # manufacturability
    },
]

product_data = [
    # ── UNITED STATES — COPPER
    {
        "region": "United States", "dominant material": "cu",
        "product": "EV Traction Motor Winding",
        "copper_kg": 5.00, "aluminum_kg": 0.00, "total_mass_kg": 6.50,
        "non_material_cost_per_unit": 48.00,          # US labor + automated hairpin tooling
        "attribute_2_value": 9.0,   # material_continuity: globally entrenched standard
```

```
    "attribute_3_value": 8.5,   # power_density: 58 MS/m; high fill factor hairpin
    "attribute_4_value": 8.0,   # thermal_performance: ~400 W/m·K; efficient heat path
    "attribute_5_value": 7.5,   # manufacturability: mature automated hairpin process
},
# ── UNITED STATES — ALUMINUM
{
    "region": "United States", "dominant material": "al",
    "product": "EV Traction Motor Winding",
    "copper_kg": 0.30, "aluminum_kg": 3.50, "total_mass_kg": 5.00,
    "non_material_cost_per_unit": 55.00,          # Al welding non-standard in US; elevate
    "attribute_2_value": 2.5,   # material_continuity: non-incumbent; major switching cost
    "attribute_3_value": 6.2,   # power_density: 35 MS/m; 65% larger cross-section required
    "attribute_4_value": 5.5,   # thermal_performance: ~200 W/m·K; inferior heat path
    "attribute_5_value": 5.0,   # manufacturability: Al end-turn welding complex; low yield
},
# ── CHINA — COPPER
{
    "region": "China", "dominant material": "cu",
    "product": "EV Traction Motor Winding",
    "copper_kg": 5.00, "aluminum_kg": 0.00, "total_mass_kg": 6.50,
    "non_material_cost_per_unit": 20.00,          # lower labor; high-volume automation
    "attribute_2_value": 9.0,
    "attribute_3_value": 8.5,
    "attribute_4_value": 8.0,
    "attribute_5_value": 7.5,
},
# ── CHINA — ALUMINUM
{
    "region": "China", "dominant material": "al",
    "product": "EV Traction Motor Winding",
    "copper_kg": 0.30, "aluminum_kg": 3.50, "total_mass_kg": 5.00,
    "non_material_cost_per_unit": 16.00,          # low labor partially offsets Al process cost
    "attribute_2_value": 4.5,   # material_continuity: most developed Al winding ecosystem
    "attribute_3_value": 5.8,   # power_density
    "attribute_4_value": 5.2,   # thermal_performance: managed via oil cooling
    "attribute_5_value": 6.0,   # manufacturability: highest Al score; advanced manufa
},
# ── INDIA — COPPER
{
    "region": "India", "dominant material": "cu",
    "product": "EV Traction Motor Winding",
    "copper_kg": 5.00, "aluminum_kg": 0.00, "total_mass_kg": 6.50,
    "non_material_cost_per_unit": 15.00,          # low labor; simpler motor platforms
    "attribute_2_value": 9.0,
```

```
        "attribute_3_value": 8.5,
        "attribute_4_value": 8.0,
        "attribute_5_value": 7.5,
    },
    # ── INDIA — ALUMINUM
    {
        "region": "India", "dominant material": "al",
        "product": "EV Traction Motor Winding",
        "copper_kg": 0.30, "aluminum_kg": 3.50, "total_mass_kg": 5.00,
        "non_material_cost_per_unit": 12.00,          # lowest NMC; low labor
        "attribute_2_value": 3.2,   # material_continuity: Al winding not established in India
        "attribute_3_value": 5.0,   # power_density: lowest Al score; lower-power applications
        "attribute_4_value": 4.5,   # thermal_performance: limited active cooling infrastructure
        "attribute_5_value": 4.5,   # manufacturability: Al welding QC a major challenge
    },
    # ── EUROPE — COPPER
    {
        "region": "Europe", "dominant material": "cu",
        "product": "EV Traction Motor Winding",
        "copper_kg": 5.00, "aluminum_kg": 0.00, "total_mass_kg": 6.50,
        "non_material_cost_per_unit": 45.00,          # EU labor + precision OEM standards
        "attribute_2_value": 9.0,
        "attribute_3_value": 8.5,
        "attribute_4_value": 8.0,
        "attribute_5_value": 7.5,
    },
    # ── EUROPE — ALUMINUM
    {
        "region": "Europe", "dominant material": "al",
        "product": "EV Traction Motor Winding",
        "copper_kg": 0.30, "aluminum_kg": 3.50, "total_mass_kg": 5.00,
        "non_material_cost_per_unit": 52.00,          # highest Al NMC; EU labor + non-standar
        "attribute_2_value": 2.8,   # material_continuity: premium OEMs strongly Cu-committed
        "attribute_3_value": 6.5,   # power_density: highest Al score; EU engineering optimization
        "attribute_4_value": 6.0,   # thermal_performance: oil-spray cooling available
        "attribute_5_value": 5.5,   # manufacturability: capable EU plants; Al winding non-standard
    },
]
```